

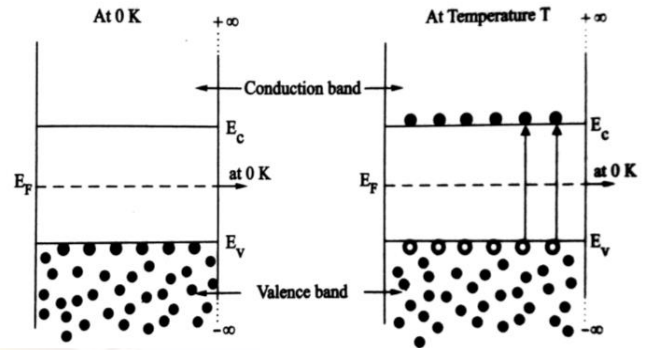
UNIT – II - SEMICONDUCTOR PHYSICS

- **Intrinsic Semiconductor**
- **Extrinsic Semiconductor (n-type & p-type semiconductor)**
- **Hall effect & Hall devices**
- **Schottky diode & Ohmic contacts**
- **Semiconducting polymer**

1. Intrinsic Semiconductor

Let us assume the mass of free electron in the conduction band is m_e^* and mass of hole in the valence band is m_h^* . Let the minimum energy corresponding to the lowest level of conduction band be E_c and the maximum energy corresponding to the highest level of valence band be E_v .

At 0 K intrinsic semiconductor behaves as insulator. But when the temperature increases some electrons move from valence band to conduction band. Therefore, both electrons in the conduction band and holes in the valence band will contribute to electrical conductivity. Therefore, the carrier concentration or density of electrons (n) and holes (p) has to be calculated.



Density of electrons in conduction band

$$\left. \begin{array}{l} \text{Density of electrons} \\ \text{in conduction band} \end{array} \right\} n = \int_{E_c}^{\infty} Z(E) \cdot F(E) dE \quad (1)$$

From Fermi-Dirac statistics we can write

$$Z(E)dE = \frac{4\pi}{h^3} (2m_e^*)^{3/2} E^{1/2} dE \quad (2)$$

Considering minimum energy of conduction band as E_c and the maximum energy can go upto ∞ , E_c denotes potential energy of an electron at rest. $(E-E_c)$ is Kinetic energy of electron at higher energy level.

$$Z(E)dE = \frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_c)^{1/2} dE \quad (3)$$

Fermi function, the probability of finding an electron in a given energy level is

$$F(E) = \frac{1}{1 + e^{(E-E_F)/K_B T}} \quad (4)$$

Substituting eqns (3) and (4) in eqn (1), we have density of electrons in conduction band is

$$\begin{aligned} n &= \int_{E_c}^{\infty} \frac{4\pi}{h^3} (2m_e^*)^{3/2} (E - E_c)^{1/2} \frac{1}{1 + e^{(E-E_F)/K_B T}} dE \\ n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2}}{1 + e^{(E-E_F)/K_B T}} dE \end{aligned} \quad (5)$$

We know E is an energy level lying in the conduction band, therefore for all possible temperatures $(E-E_F) \gg K_B T$ (or) $(E-E_F)/K_B T \gg 1$ (or) $e^{(E-E_F)/K_B T} \gg 1$

$$\therefore 1 + e^{(E-E_F)/K_B T} \approx e^{(E-E_F)/K_B T}$$

Therefore, equation (5) becomes

$$\begin{aligned}
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} \frac{(E - E_c)^{1/2}}{e^{(E-E_F)/K_B T}} dE \\
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-(E-E_F)/K_B T} dE \\
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{(E_F-E)/K_B T} dE \\
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{E_F/K_B T} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-E/K_B T} dE \quad (6)
\end{aligned}$$

To evaluate the integral in eqn (6), let us substitute

$E - E_c = x$ (or) $E = E_c + x$ & $dE = dx$

Limits: when $E = E_c$; $x = 0$

when $E = \infty$; $x = \infty$

Therefore, the limits are 0 to ∞ , Now equation (6) becomes

$$\begin{aligned}
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{E_F/K_B T} \int_0^{\infty} x^{1/2} e^{-(E_c+x)/K_B T} dx \\
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{(E_F-E_c)/K_B T} \int_0^{\infty} x^{1/2} e^{-x/K_B T} dx \quad (7)
\end{aligned}$$

Using the gamma function, we can get

$$\int_0^{\infty} x^{1/2} e^{-x/K_B T} dx = \frac{(K_B T)^{3/2} \pi^{1/2}}{2} \quad (8)$$

Substituting eqn (8) in eqn (7), we can get

$$\begin{aligned}
n &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} e^{(E_F-E_c)/K_B T} \left[\frac{(K_B T)^{3/2} \pi^{1/2}}{2} \right] \\
n &= \frac{2\pi}{h^3} (2m_e^*)^{3/2} (K_B T)^{3/2} \pi^{1/2} e^{(E_F-E_c)/K_B T} \\
n &= \frac{2\pi \pi^{1/2} (2m_e^*)^{3/2} (K_B T)^{3/2} e^{(E_F-E_c)/K_B T}}{(h^2)^{3/2}}
\end{aligned}$$

$\therefore$ Density of electrons in conduction band is

$$n = 2 \left[\frac{2\pi m_e^* K_B T}{h^2} \right]^{3/2} e^{(E_F-E_c)/K_B T} \quad (9)$$

The above equation represents the equation for density of electrons.

Density of holes in valence band

$F(E)$ represents the probability of filled state or probability of finding an electron in the given state. As the maximum probability is 1, the probability of unfilled state or the probability of finding a hole will be $[1-F(E)]$.

We know, the maximum energy in the valence band is E_v and the minimum energy is $-\infty$. Therefore, the density of holes in valence band p is given by

$$\left. \begin{array}{l} \text{Density of holes} \\ \text{in valence band} \end{array} \right\} p = \int_{-\infty}^{E_v} Z(E) \cdot [1 - F(E)] dE \quad (1)$$

The probability of finding a hole in a given energy level is

$$\begin{aligned} 1 - F(E) &= 1 - \frac{1}{1 + e^{(E-E_F)/K_B T}} \\ &= \frac{1 + e^{(E-E_F)/K_B T} - 1}{1 + e^{(E-E_F)/K_B T}} \\ &= \frac{e^{(E-E_F)/K_B T}}{1 + e^{(E-E_F)/K_B T}} \end{aligned} \quad (2)$$

We know E is an energy level lying in the valence band, therefore for all possible temperatures $E-E_F \ll K_B T$ (or) $E-E_F / K_B T \ll 1$ (or) $e^{(E-E_F)/K_B T} \ll 1$

$$\begin{aligned} 1 + e^{(E-E_F)/K_B T} &\approx 1 \\ \therefore 1 - F(E) &= e^{(E-E_F)/K_B T} \end{aligned} \quad (3)$$

Density of states in valence band

$$Z(E)dE = \frac{4\pi}{h^3} (2m_h^*)^{3/2} E^{1/2} dE \quad (4)$$

E_v , top of energy level in valence band and is the potential energy of a hole at rest. (E_v-E) is kinetic energy of the hole at level below E_v . Hence E in eqn (4) is replaced by (E_v-E)

$$Z(E)dE = \frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_v - E)^{1/2} dE \quad (5)$$

Substituting eqns(3) and (5) in (1)

$$dp = \frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_v - E)^{1/2} e^{(E-E_F)/K_B T} dE \quad (6)$$

The number of holes in valence band for the entire energy range is obtained by integrating eqn(6) within the limits $-\infty$ to E_v

$$\begin{aligned} \int dp &= p = \int_{-\infty}^{E_v} \frac{4\pi}{h^3} (2m_h^*)^{3/2} (E_v - E)^{1/2} e^{(E-E_F)/K_B T} dE \\ p &= \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(-E_F)/K_B T} \int_{-\infty}^{E_v} (E_v - E)^{1/2} e^{(E)/K_B T} dE \end{aligned} \quad (7)$$

Let us substitute $E_v - E = x$ (or) $E = E_v - x$ and $dE = -dx$

Limits: when $E = -\infty$; $E_v - (-\infty) = x$; $x = \infty$

when $E = E_v$; $x = 0$

Therefore, the limits are ∞ to 0

Now equation (7) becomes

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(-E_F)/K_B T} \int_{\infty}^0 x^{1/2} e^{(E_v-x)/K_B T} (-dx) \quad (8)$$

To exclude the negative sign, we can interchange the limits.

$$p = \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(E_v - E_F)/K_B T} \int_0^\infty x^{1/2} e^{-x/K_B T} dx \quad (9)$$

Using the gamma function, we can get

$$\int_0^\infty x^{1/2} e^{-x/K_B T} dx = \frac{(K_B T)^{3/2} \pi^{1/2}}{2} \quad (10)$$

Substituting eqn (10) in eqn (9), we can get

$$\begin{aligned} p &= \frac{4\pi}{h^3} (2m_h^*)^{3/2} e^{(E_v - E_F)/K_B T} \left[\frac{(K_B T)^{3/2} \pi^{1/2}}{2} \right] \\ p &= \frac{2\pi}{h^3} (2m_h^*)^{3/2} (K_B T)^{3/2} \pi^{1/2} e^{(E_v - E_F)/K_B T} \\ p &= \frac{2\pi \pi^{1/2} (2m_h^*)^{3/2} (K_B T)^{3/2} e^{(E_v - E_F)/K_B T}}{(h^2)^{3/2}} \end{aligned}$$

∴ Density of holes in valence band is

$$p = 2 \left[\frac{2\pi m_h^* K_B T}{h^2} \right]^{3/2} e^{(E_v - E_F)/K_B T} \quad (11)$$

The above equation represents the equation for density of holes.

Intrinsic Carrier Concentration

In intrinsic semiconductor, the number of electrons in conduction band is equal to the number of holes in valence band.

$$n_i = n = p \quad (1)$$

$$n_i \times n_i = n_i^2 = np \quad (2)$$

$$n_i^2 = 2 \left[\frac{2\pi m_e^* K_B T}{h^2} \right]^{3/2} e^{(E_F - E_c)/K_B T} \times 2 \left[\frac{2\pi m_h^* K_B T}{h^2} \right]^{3/2} e^{(E_v - E_F)/K_B T}$$

$$n_i^2 = 4 \left[\frac{2\pi K_B T}{h^2} \right]^3 (m_e^* m_h^*)^{3/2} e^{(E_v - E_c)/K_B T}$$

$$n_i^2 = 4 \left[\frac{2\pi K_B T}{h^2} \right]^3 (m_e^* m_h^*)^{3/2} e^{-E_g / K_B T} \quad (3)$$

where $E_c - E_v = E_g$ is energy gap.

Taking square root on both sides in eqn(3)

$$(n_i^2)^{1/2} = \left(4 \left[\frac{2\pi K_B T}{h^2} \right]^3 (m_e^* m_h^*)^{3/2} e^{-E_g / K_B T} \right)^{1/2}$$

$$n_i = 2 \left[\frac{2\pi K_B T}{h^2} \right]^{3/2} (m_e^* m_h^*)^{3/4} e^{-E_g / 2K_B T}$$

Variation of Fermi level with temperature in an intrinsic semiconductor

For an intrinsic semiconductor, the number of electrons (n) is equal to the number of holes (p). Equating expression for n and p

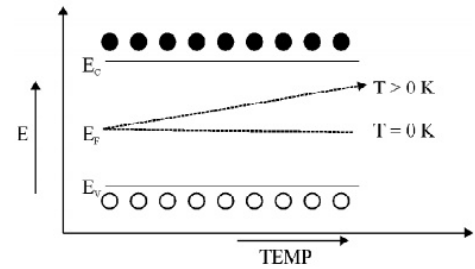
$$2 \left[\frac{2\pi m_e^* K_B T}{h^2} \right]^{3/2} e^{(E_F - E_c) / K_B T} = 2 \left[\frac{2\pi m_h^* K_B T}{h^2} \right]^{3/2} e^{(E_v - E_F) / K_B T} \quad (1)$$

$$(m_e^*)^{3/2} e^{(E_F - E_c) / K_B T} = (m_h^*)^{3/2} e^{(E_v - E_F) / K_B T}$$

$$\left(\frac{m_h^*}{m_e^*} \right)^{3/2} = \frac{e^{(E_F - E_c) / K_B T}}{e^{(E_v - E_F) / K_B T}}$$

$$= e^{(E_F - E_c - E_v + E_F) / K_B T}$$

$$\left(\frac{m_h^*}{m_e^*} \right)^{3/2} = e^{(2E_F - (E_v + E_c)) / K_B T}$$



Taking log on both sides,

$$3/2 \log_e \left(\frac{m_h^*}{m_e^*} \right) = \frac{2E_F - (E_v + E_c)}{K_B T}$$

$$2E_F = E_c + E_v + 3/2 K_B T \log_e \frac{m_h^*}{m_e^*}$$

$$E_F = \frac{E_c + E_v}{2} + \frac{3}{4} K_B T \log_e \left(\frac{m_h^*}{m_e^*} \right) \quad (2)$$

If $m_h^* = m_e^*$ then $\log_e \frac{m_h^*}{m_e^*} = 0$ [$\because \log_e 1 = 0$]

Equation (2) becomes

$$E_F = \frac{(E_c + E_v)}{2} \quad (3)$$

From equation (3), the Fermi energy level lies in the midway between E_c and E_v in an intrinsic semiconductor at absolute zero. But $m_h^* > m_e^*$ and the Fermi energy level slightly increases with the increase in temperature.

2. Determination of band gap energy

We know for intrinsic semiconductor $\sigma_i = C e^{-E_g / 2K_B T}$ where C is a constant.

We know resistivity $\rho_i = 1 / \text{conductivity}$

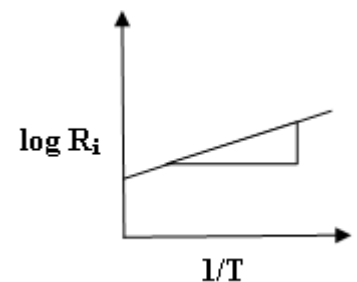
$$\rho_i = \frac{1}{C} e^{E_g / 2K_B T} \quad (1)$$

We know resistivity is resistance per unit area per unit length

$$\rho_i = \frac{R_i a}{l} \quad (2)$$

Equating equations (1) and (2) we get,

$$\frac{R_i a}{l} = \frac{1}{C} e^{E_g / 2K_B T} \quad (\text{or}) \quad R_i = \frac{1}{aC} e^{E_g / 2K_B T}$$



(or)

$$R_i = C_1 e^{E_g / 2K_B T} \quad \text{where } C_1 = \frac{1}{aC}$$

Taking log on both sides we can write $\log R_i = \log C_1 + \frac{E_g}{2K_B T}$

The resistance can be found using Meter bridge or Carey fosters bridge or Post office box experiments at various temperatures. If a graph is plotted between $1/T$ and $\log R_i$ with a straight line is obtained as shown in fig with a slope.

$$\frac{dy}{dx} = \frac{E_g}{2KB}$$

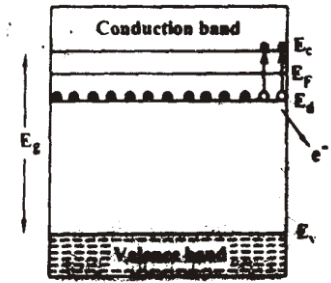
Therefore by finding the slop of line we can calculate the energy band gap E_g with the following

expression: $E_g = 2KB \frac{(dy)}{(dx)} \text{ Joules}$

3. Carrier concentration in n-type semiconductor

The energy band structure of n-type semiconductor is as shown in figure. At 0 K, E_F will lie exactly between E_c and E_d , but even at low temperature some electrons may go from E_d to E_c . Let us assume that $E_c - E_F > K_B T$ Then the density of electrons in conduction band can be written as

$$n = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} e^{(E_F - E_c) / K_B T} \quad (1)$$



Let N_d be the number of donor energy levels per cm^3 (i.e) density of state $Z(E_d)dE$, which has energy E_d below the conduction band.

If some electrons are donated from the donor energy level to conduction band say for example if two electrons go to conduction band then two vacant sites (holes) will be created in E_d levels as shown in figure.

Thus, in general we can write the density of holes in donor energy level as

$$= N_d \left[1 - \frac{1}{1 + e^{(E_d - E_F) / K_B T}} \right] \quad (2)$$

$$\begin{aligned} &= N_d \left[\frac{1 + e^{(E_d - E_F) / K_B T}}{1 + e^{(E_d - E_F) / K_B T}} \right] \\ &= \left[\frac{N_d e^{(E_d - E_F) / K_B T}}{1 + e^{(E_d - E_F) / K_B T}} \right] \end{aligned} \quad (3)$$

$e^{(E_d - E_F) / K_B T}$ is very small in eqn(3) when compared to 1. We can neglect 1 from denominator.

$$1 + e^{(E_d - E_F) / K_B T} \approx 1$$

$$\text{Density of ionized donor} = N_d e^{(E_d - E_F) / K_B T} \quad (4)$$

At equilibrium,

Density of electron in conduction band = Density of ionized donors

$$n = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} e^{(E_F - E_c) / K_B T} = N_d e^{(E_d - E_F) / K_B T} \quad (5)$$

$$\frac{e^{(E_F - E_c) / K_B T}}{e^{(E_d - E_F) / K_B T}} = \frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}}$$

$$e^{(E_F - E_c) / K_B T} e^{-(E_d - E_F) / K_B T} = \frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}}$$

$$e^{(E_F - E_c - E_d + E_F) / K_B T} = \frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \quad (6)$$

Taking log on both sides

$$\log_e e^{(E_F - E_c - E_d + E_F) / K_B T} = \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right],$$

$$\frac{E_F - E_c - E_d + E_F}{K_B T} = \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right]$$

$$2E_F - (E_c + E_d) = K_B T \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right]$$

$$(\text{or}) \quad 2E_F = (E_d + E_c) + K_B T \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right]$$

$$E_F = \frac{(E_d + E_c)}{2} + K_B T \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right] \quad (7)$$

Substituting the expression of E_F from (7) in (1)

$$n = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} \exp \left[\frac{\left(\frac{E_d + E_c}{2} + K_B T \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right] - E_c \right)}{K_B T} \right] \quad (8)$$

$$n = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} \exp \left[\left[\frac{(E_d + E_c - 2E_c)}{2K_B T} \right] + \frac{1}{2} \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right] \right]$$

$$n = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} \exp \left[\left[\frac{(E_d - E_c)}{2K_B T} \right] + \log_e \left[\frac{N_d^{1/2}}{\left(2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} \right)^{1/2}} \right] \right]$$

$$n = 2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} e^{(E_d - E_c)/2K_B T} e^{\log_e \left[\frac{N_d^{1/2}}{2^{1/2} \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/4}} \right]} \quad (9)$$

Rearranging the expression (9)

$$n = 2 \frac{N_d^{1/2}}{2^{1/2}} \frac{\left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}}{\left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/4}} e^{(E_d - E_c)/2K_B T}$$

$$n = 2^{1/2} 2^{1/2} \frac{N_d^{1/2}}{2^{1/2}} \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2} \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{-3/4} e^{(E_d - E_c)/2K_B T}$$

$$n = 2^{1/2} N_d^{1/2} \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/4} e^{(E_d - E_c)/2K_B T} \quad (10)$$

$$n = (2N_d)^{1/2} \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/4} e^{(-\Delta E)/2K_B T} \quad (11)$$

where $\Delta E = E_c - E_d$

The above expression is the charge carrier concentration for n-type semiconducting material.

Variation of Fermi level with temperature and donor impurity concentration

Fermi level for n-type semiconducting material is as follows

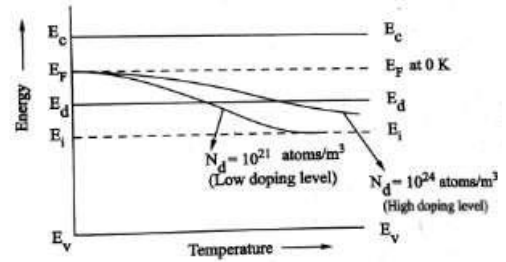
$$E_F = \frac{E_c + E_d}{2} + \frac{K_B T}{2} \log_e \left[\frac{N_d}{2 \left(\frac{2\pi m_e^* K_B T}{h^2} \right)^{3/2}} \right]$$

At 0 K i.e when $T = 0$, we can write the above equation as

$$E_F = \frac{E_c + E_d}{2}$$

The Fermi level decreases with increase in temperature as all the impurity atoms get ionized as temperature increases. At high temperature the material behave as an intrinsic semiconductor. For an intrinsic semiconductor the Fermi level lies near the middle of the band gap so the Fermi level decreases with temperature.

As the impurity concentration increases the Fermi level increases as the level for an n-type material exist near the middle of impurity energy level and conduction band.

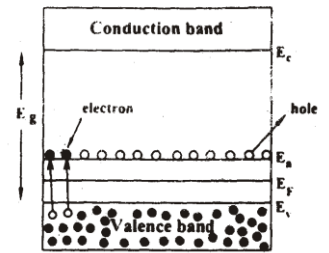


4. Carrier concentration in 'p' type semiconductor

For p-type at absolute zero E_F will be exactly between E_a and E_v . At low temperatures some electrons from valence band fills the holes in the acceptor energy levels as shown in fig. We know the density of holes in the valence band,

$$p = 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} e^{(E_v - E_F) / K_B T} \quad (1)$$

Let N_a be the number of acceptor energy levels per cm^3 which has energy E_a above valence band. If some electrons are accepted by the acceptor energy levels from the valence band, say for example, if two electrons are accepted to fill the hole sites in the acceptor levels, then two holes will be created in the valence band as shown in fig.



Therefore, in general, the electron density in the acceptor energy level can be written as

$$\text{Density of ionized acceptors} = N_a F(E_a) \quad (2)$$

$$F(E_a) = \frac{1}{1 + e^{(E_a - E_F) / K_B T}}$$

where, E_a - acceptor energy level

$F(E_a)$ - probability for finding electron in acceptor energy level (Ionized acceptor)

$$F(E_a) = \frac{N_a}{1 + e^{(E_a - E_F) / K_B T}} \quad (3)$$

$e^{(E_a - E_F) / K_B T}$ is large quantity and hence 1 can be neglected from eqn (3)

$$N_a F(E_a) = \frac{N_a}{e^{(E_a - E_F) / K_B T}}$$

$$N_a F(E_a) = N_a e^{-(E_a - E_F) / K_B T}$$

$$\text{Density of ionized acceptors} = N_a e^{(E_F - E_a) / K_B T} \quad (4)$$

At equilibrium condition,

Number of holes per unit Volume in valence band (i.e hole density) = Number of electrons per unit volume in acceptor energy level (i.e electron density)

Equating equation (1) and (4) we have

$$2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} e^{(E_v - E_F) / K_B T} = N_a e^{(E_F - E_a) / K_B T} \quad (5)$$

Rearranging eqn (5)

$$\frac{e^{(E_v - E_F) / K_B T}}{e^{(E_F - E_a) / K_B T}} = \frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}}$$

$$e^{(E_v - E_F) / K_B T} e^{-(E_F - E_a) / K_B T} = \frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}}$$

$$e^{(E_v - E_F - E_F + E_a) / K_B T} = \frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \quad (6)$$

Taking log on both sides

$$\log_e e^{(E_v - E_F - E_F + E_a) / K_B T} = \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]$$

$$\frac{E_v - 2E_F + E_a}{K_B T} = \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]$$

$$E_a + E_v - 2E_F = K_B T \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right] \quad (7)$$

Rearranging above eqn

$$2E_F = E_a + E_v - K_B T \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]$$

$$E_F = \frac{E_a + E_v}{2} - \frac{K_B T}{2} \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right] \quad (8)$$

Substituting eqn (8) in eqn (1)

$$\begin{aligned}
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} \exp \left[\frac{E_v - \left[\frac{(E_a + E_v)}{2} \right] - \frac{K_B T}{2} \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]}{K_B T} \right] \\
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} \exp \left[\frac{E_v - \left[\frac{(E_a + E_v)}{2} \right] + \frac{K_B T}{2} \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]}{K_B T} \right] \quad (9) \\
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} \exp \left[\frac{2E_v - E_v - E_a}{2K_B T} + \frac{1}{2} \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right] \right] \\
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} \exp \left[\frac{E_v - E_a}{2K_B T} + \log_e \left[\frac{\frac{N_a}{2}}{\left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]^{\frac{1}{2}} \right] \\
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} \exp \left[\frac{E_v - E_a}{2K_B T} + \log_e \left[\frac{\left(\frac{N_a}{2} \right)^{1/2}}{\left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/4}} \right] \right] \\
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} e^{(E_v - E_a)/K_B T} e^{\log_e \left[\frac{\left(\frac{N_a}{2} \right)^{1/2}}{\left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/4}} \right]} \\
p &= 2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} e^{(E_v - E_a)/K_B T} \frac{\left(\frac{N_a}{2} \right)^{1/2}}{\left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/4}}
\end{aligned}$$

$$p = 2 \frac{N_a^{1/2}}{2^{1/2}} \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} e^{(E_v - E_a)/K_B T}$$

$$p = 2 \frac{N_a^{1/2}}{2^{1/2}} \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2} \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/4} e^{(E_v - E_a)/K_B T} \quad (9)$$

$$p = (2N_a)^{1/2} \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/4} e^{-\Delta E/K_B T} \quad (10)$$

The above expression is the charge carrier concentration for p-type semiconducting material.

Variation of Fermi level with temperature and acceptor impurity concentration

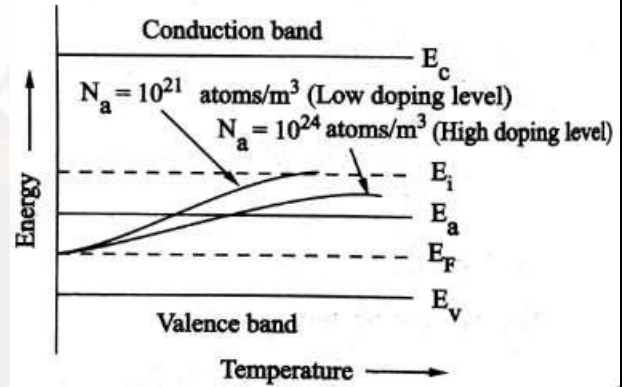
$$E_F = \frac{E_a + E_v}{2} - \frac{K_B T}{2} \log_e \left[\frac{N_a}{2 \left(\frac{2\pi m_h^* K_B T}{h^2} \right)^{3/2}} \right]$$

At 0 K i.e when $T = 0$ K, we can write equation (8) as

$$E_F = \frac{E_a + E_v}{2}$$

The Fermi level increases with increase in temperature as all the impurity atoms get ionized as temperature increases. At high temperature the material behaves as an intrinsic semiconductor. For an intrinsic semiconductor the Fermi level lies near the middle of the band gap so the Fermi level increases with temperature.

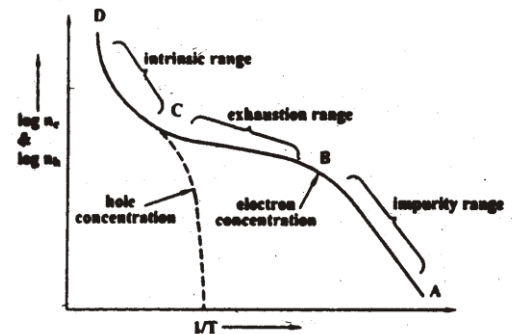
As the impurity concentration increases the Fermi level decreases as the level for p-type material exist near the middle of impurity energy level and valence band.



5.The carrier concentration and electrical conductivity vary with temperature in an extrinsic semiconductor

In extrinsic semiconductor, the resistivity decreases linearly with increase in temperature. This variation is considered over three different regions, (i) Extrinsic or impurity range (ii) Exhaustion range (iii) Intrinsic range. For an n-type semiconductor, the variation of carrier concentration n and p with temperature is shown in diagram. At 0 K, both conduction and valence bands are free from any charge carriers and hence, the electrical conductivity is zero.

With increase in temperature, the donor atoms get ionized and hence electron concentration in conduction band increases with temperature until all the donor atoms are ionized. This range is known as impurity or extrinsic range. When the temperature is further increased to room temperature, there are no more donor atoms to be ionized and so the concentration of electrons in conduction band remains constant over a certain

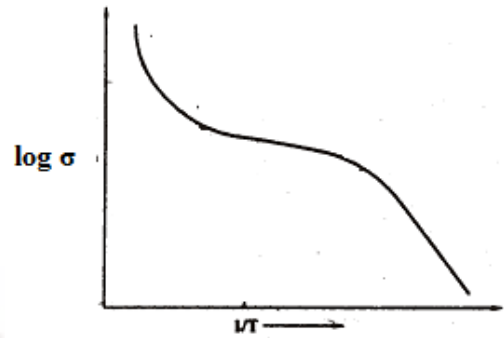


temperature range. This region is known as exhaustion range. As the temperature is increased further, the electrons in valence band are lifted across the forbidden gap to conduction band. Thus, electron concentration increases in conduction band considerably.

With further increase in temperature, more and more electrons from valence band reach the conduction band and completely outnumber the donor electrons. The material practically becomes intrinsic and so this range is called as intrinsic range. All these ranges are shown in diagram. The dotted curve in figure indicates the hole concentration in an intrinsic range.

For p-type

The difference between the p-type and n-type material is that the initial increase in the carrier concentration in the extrinsic range is due to the hole concentration since the majority charge carriers in p-type materials are holes. Even in p-type material the exhaustion range exists where there is no change in charge carriers for few temperatures range and it increases drastically at high temperature when the electrons from valence band are excited to the conduction band. In an intrinsic region both p-type and n-type material behave in the same manner.



Variation of electrical conductivity in extrinsic material at low temperature i.e. near absolute zero, only few impurity atoms are ionized and hence the electrical conductivity is low. With the increase of temperature, more and more of impurity atoms get ionized and so electrical conductivity increases. The variation of electrical conductivity with temperature in extrinsic semiconductor is shown in diagram.

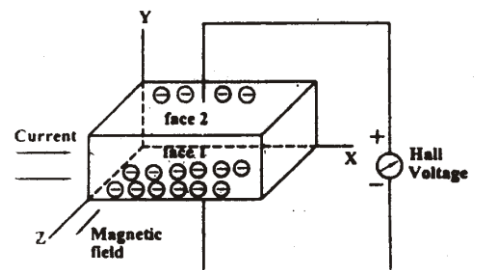
At higher temperatures, due to the generation of electron-hole pairs, the electrical conductivities increase steeply and the material becomes practically intrinsic.

6. Hall Effect

When a conductor (metal or semiconductor) carrying a current is placed in a transverse magnetic field, an electric field is produced inside the conductor in a direction normal to both the current and the magnetic field. This phenomenon is known as “Hall Effect” and the generated voltage is called the “Hall voltage”.

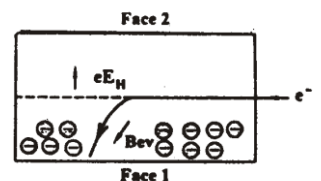
Hall Effect in n-type Semiconductor

Let us consider an n-type material to which the current is allowed to pass along x-direction from left to right and the magnetic field is applied in z-direction. As a result, Hall voltage is produced in y direction as shown in figure



Since the direction of current is from left to right the electrons move from right to left in X-direction as shown in figure.

Now due to the magnetic field applied the electrons move towards downward direction with the velocity ‘v’ and cause the negative charge to accumulate at face (1) of the material as shown in figure. Therefore, a potential difference is established between face (2) and face (1) of the specimen which gives rise to field E_H in the negative y direction.



Here, the Force due to potential difference = eE_H (1)

Force due to magnetic field = Bev (2)

At equilibrium equation (1) = equation (2)

$$\begin{aligned} eE_H &= Bev \\ E_H &= Bv \end{aligned} \quad (3)$$

We know the current density J_x in the X direction is

$$\begin{aligned} J_x &= -n_e ev; \\ v &= \frac{-J_x}{n_e e} \end{aligned} \quad (4)$$

Substituting equation (4) in equation (3) we get

$$\begin{aligned} E_H &= \frac{-BJ_x}{n_e e} \\ E_H &= R_H J_x B \end{aligned} \quad (5)$$

where R_H is known as the Hall coefficient, given by

$$R_H = -1/n_e e$$

The negative sign indicates that the field is developed in the negative Y direction.

Hall Coefficient in terms of Hall voltage

If the thickness of the sample is t and the voltage developed is V_H then

$$\text{Hall voltage } V_H = E_H t \quad (6)$$

Substituting equation (5) in equation (6) we have

$$V_H = R_H J_x B t \quad (7)$$

If b is the width

Substituting equation (8) in equation (7) we get

$$V_H = \frac{R_H I_x B t}{bt}; \quad V_H = \frac{R_H I_x B}{b}$$

$$\text{Hall coefficient } R_H = \frac{V_H b}{I_x B}$$

Note: The sign for V_H will be opposite for n and p type semiconductors.

Experimental setup for the measurement of Hall voltage

A semiconductor slab of thickness ' t ' and breadth ' b ' is taken and current is passed using the battery as shown in fig.

The slab is placed between the pole pieces of an electromagnet so that current direction coincides with X-axis and magnetic field coincides with Z-axis. The Hall voltage (V_H) is measured by placing two probes at the centre of the top and bottom faces of the slab (Y-axis)

If B is the magnetic field applied and V_H is the Hall voltage produced, then the Hall coefficient can be calculated from the formula

$$R_H = \frac{V_H b}{I_x B}$$

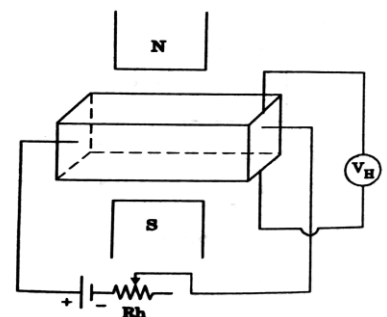
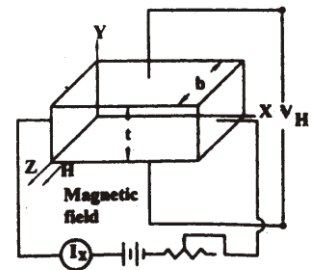
7. HALL DEVICES

The device which uses the Hall Effect for its application is known as Hall device. There are three types of Hall devices.

- Gauss Meter
- Electronic Multiplier
- Electronic Wattmeter

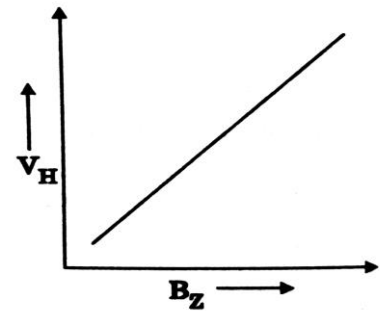
a) Gauss Meter

The Hall voltage $V_H = R_H B_Z I_X / t$. In this, $V_H \propto B_Z$ for a



given hall element; R_H and t are constant.

The current (I) through Hall element is also kept constant. This principle is used in Gauss meter. It is used for measuring magnetic field. The variation of Hall voltage with magnetic field is shown in fig. The voltmeter which is used to measure V_H can be directly calibrated in terms of Gauss. The graph can be also used to measure any unknown magnetic fields.



b) Electronic Multipliers

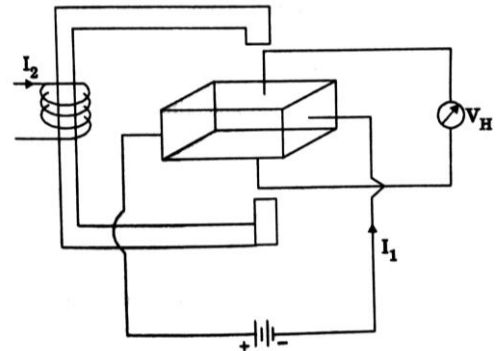
From Hall effect, we have $V_H = R_H B_z I_x / t$
Since R_H and t are constant for an element

$$V_H \propto B_z I_1$$

But, the magnetic field B is proportional to current (I_2) through the coil.

$$V_H \propto I_1 I_2$$

i.e., V_H is a measure of the product of two currents. This is the basic principle used in analog electronic multipliers. The fig. shows the circuit diagram for electronic multiplier.



c) Electronic Wattmeter

Hall effect is used to measure electrical power dissipated in a load. The instrument used to measure the power in a circuit using Hall effect principle is known as Hall effect - Watt meter.

S is Hall Effect sample. It is placed in a magnetic field B_z produced by the load current I_L passing through the coils CC as shown in fig.

V_L - Load voltage,

I_L - Load Current C ,

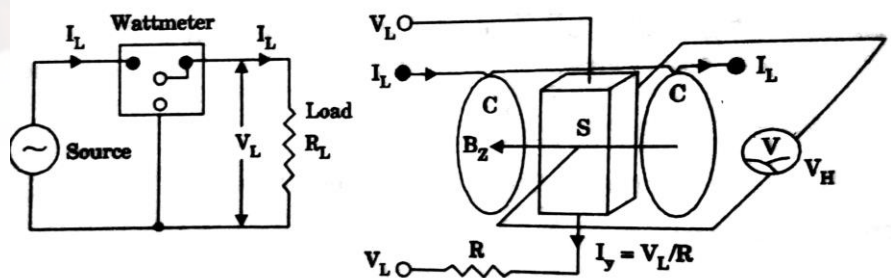
C - Coils to set magnetic field B

The voltage across the load V drives the current through the sample. R is a series resistance which is greater than the resistance of the sample and that of the load. If ' t ' thickness of the sample,

then the measured Hall voltage $V_H = \frac{R_H B_z I_y}{t}$

$$V_H = I_L V_L$$

This is the electric power dissipated by the load. The voltmeter that measures V_H can be calibrated to read power directly.

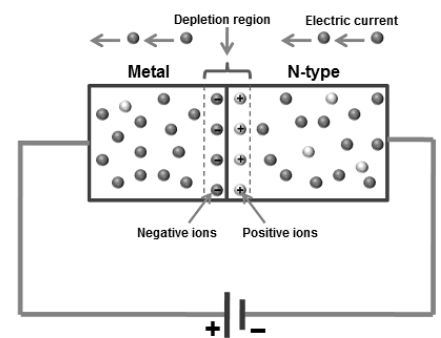


8. Schottky Diode

Definition

When the metal has a higher work function than that of n-type semiconductor then the junction is called schottky diode.

The electrons in the conduction level of the semiconductor



Physics and Radio-Electronics

Forward biased schottky diode

move to the empty energy states above the Fermi level of the metal. This leaves a positive charge on the semiconductor side and a negative charge on the metal side. This leads to a contact potential.

There is a built in potential V_o in the Schottky junction. This is given by the difference in work functions.

$$eV_o = \phi_m - \phi_{semi}$$

Forward-Biased State

Connecting the positive terminal of a battery to the metal and negative terminal to the n-type semiconductor will create a forward-biased state. In this state, electrons can cross the junction from n-type to metal if the applied voltage is greater than 0.2 volts. This results in a flow of current that's typical for most diodes.

Reverse-Biased State

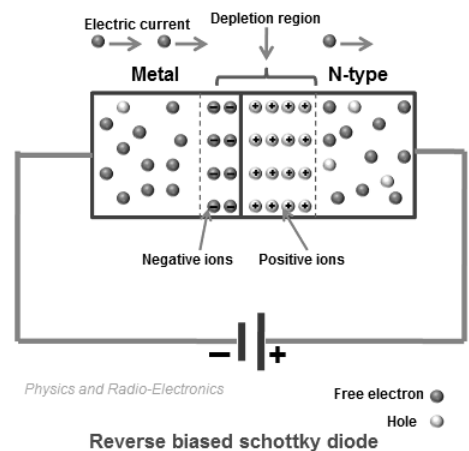
Connecting the negative terminal of a diode to the metal and positive terminal to the n-type semiconductor will create a reverse-biased state. This state expands the Schottky diode and prevents the flow of electric current. However, if the reverse bias voltage continues to increase this can eventually break down the diode. Doing so will allow current to flow in the reverse direction and may damage the component.

Advantages of Schottky Diode

Low turn on voltage: The turn on voltage for the diode is between 0.2 and 0.3 volts. For a silicon diode it is against 0.6 to 0.7 volts from a standard silicon diode.

Fast recovery time: A fast recovery time means a small amount of stored charge that can be used for high speed switching applications.

Low junction capacitance: It occupies a very small area, after the result obtained from wire point contact of the silicon. Since the capacitance levels are very small.



9. Ohmic Contacts

Definition

An ohmic contact is a type of metal semiconductor junction. It is formed by a contact of a metal with a heavily doped semiconductor. When the semiconductor has a higher work function than that of metal, then the junction formed is called the ohmic junction.

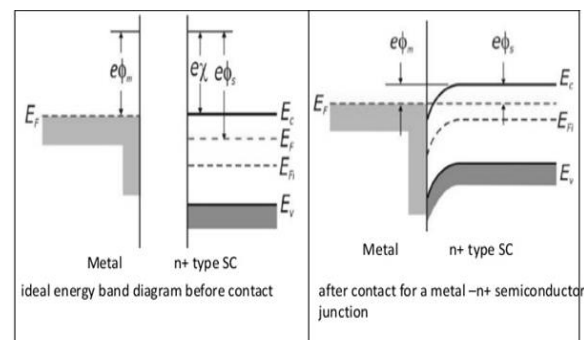
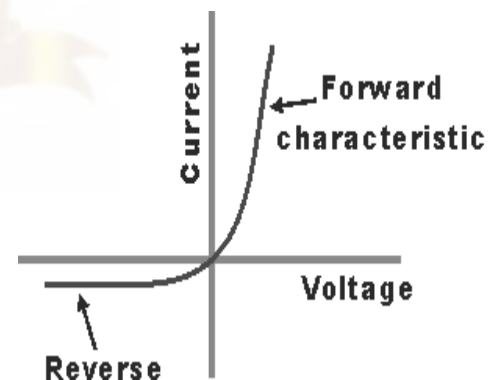
Working

Before contact, Fermi levels of the metal and semiconductor are at different positions.

At equilibrium, the electrons move from the metal to the empty states in the conduction band of semiconductor. Thus, there is an accumulation region near the interface. The accumulation region has a higher conductivity than the bulk semiconductor due to this higher concentration of electrons. Thus, a ohmic contact behaves as a resistor conducting in both forward and reverse bias. The resistivity is determined by the bulk resistivity of the semiconductor.

V-I Characteristics

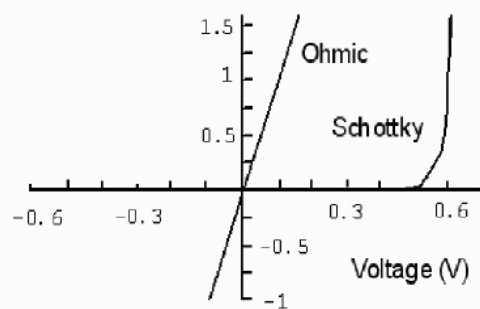
The current is directly proportional to the



potential across the junction and it is symmetric about the origin. Thus, Ohmic contacts are non-rectifying and show negligible voltage drop and resistance irrespective of the direction and magnitude of current.

Application

The use of ohmic contacts is to connect one semiconductor device to another, an IC, or to connect and IC to its external terminals.



10. SEMICONDUCTING POLYMERS

Semi-conducting polymers are having the energy gap between the valence band and conduction band (band gap) are not so large and not so small. They have low conductivity in general due to this low conductivity these are not used in practical field.

The conductivity of these semiconductor however enhance by many ways

1. Doping

2. Filled with conducting polymer

3. Blending with conducting polymer

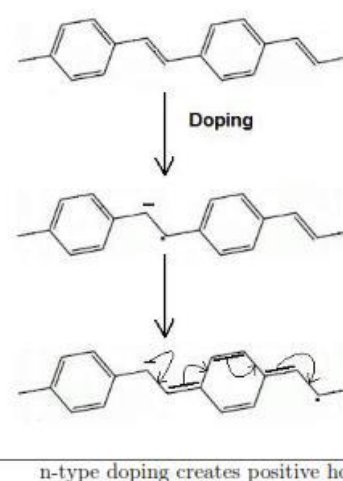
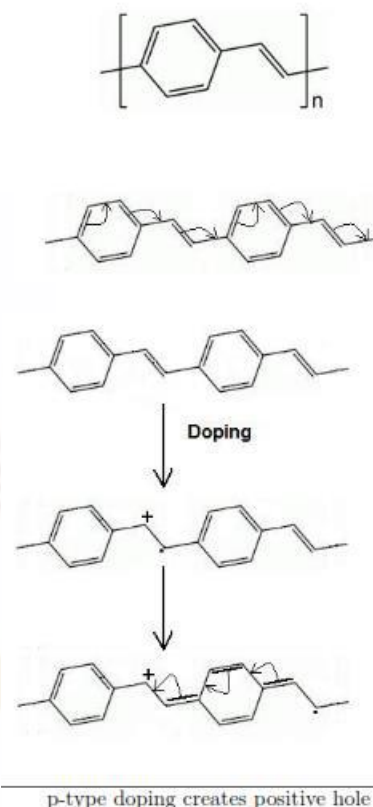
1.Doping: The conduction power of semiconductor can be enhanced by input some foreign material or impurities. These impurities are called doping agent or dopant. Appropriate doping agent increase the conductivity of semiconductors upto 10^4 times. Doping enhances the conductivity of semiconductor and called extrinsic semiconductor. The increase in conduction is due to participation of impurity elements in between the valence band and conduction band and thus making a bridge through which electrons can jump easily from the valence band to the conduction band. Actually the conjugated π electrons have very low ionization potential and high electron affinities the foreign materials develop positive or negative charge through oxidation or reduction of the semiconductor.

Doping is mainly two types.

(i) p-type doping through oxidation of materials: In this type of doping some electrons from the conjugated π bonds are removed through oxidation creating a positive hole called polaron inside the polymer. The positive hole or polaron can move throughout the polymeric chain and make it conducting polymer. Lewis acids (FeCl_3 , AlCl_3) are generally use as doping agent.

(ii) n-type through reduction of materials: In this type of doping some electrons are introduced to the conjugated π bonds through reduction creating a negative hole or charge inside the polymer. The negative hole or charge can move throughout the polymeric chain and make it conducting polymer.

2. Filled with conducting polymer: The conduction power of semiconductor can be enhanced by input some foreign conducting material or good conductor in powder (carbon dust) form or granule from (metallic fibers). The role of



polymer is to bind the conducting materials. The minimum concentration of conducting filler required to start the conduction is called percolation threshold.

3. Blending with conducting polymer: The conduction power of semiconductor can be enhanced by input some foreign conducting polymer material. During such type of blending we should alert about the compatibility of those polymer. In such cases the strength properties of the blend polymer also enhanced.

Applications of conducting polymer

Due to light weight and low cost than metal semiconductor and easy to preparation conducting polymers have diversified application.

- Rechargeable light weight batteries
- Optical display device
- Solar cell
- Telecommunication system
- Electronic toys
- As analytical sensor.

